A+ LAB SERIES V5

Lab 7: Linux Network Settings

Document Version: 2026-01-02

Material in this Lab Aligns to the Following	
CompTIA A+ (220-1201) Exam Objectives	2.4: Explain common network configuration concepts. 2.3: Summarize services provided by networked hosts. 2.6: Given a scenario, configure basic wired/wireless small office/home office (SOHO) networks. 2.7: Compare and contrast internet connection types, network types, and their characteristics.
CompTIA A+ (220-1202) Exam Objectives	1.9: Identify common features and tools of the Linux client/desktop operating system. 2.7: Given a scenario, apply workstation security options and hardening techniques. 4.9: Given a scenario, use remote access technologies.

Contents

Introduction

In this laboratory exercise, participants will delve into the practical aspects of network management on Linux and draw parallels with contemporary Windows operating systems, specifically Windows 10 and 11. The lab is structured to provide hands-on experience in configuring network settings, implementing file-sharing solutions, and utilizing a suite of command-line tools. Through this exploration, participants will not only enhance their proficiency in Linux networking but also gain insights into the comparative network management techniques between Linux and Windows environments. This comparative approach will foster a deeper understanding of network configuration and troubleshooting, equipping participants with the skills needed to operate effectively across diverse operating systems. The outlined objectives serve as a roadmap, guiding participants through each segment of the lab and ensuring a thorough grasp of networking fundamentals on Linux.

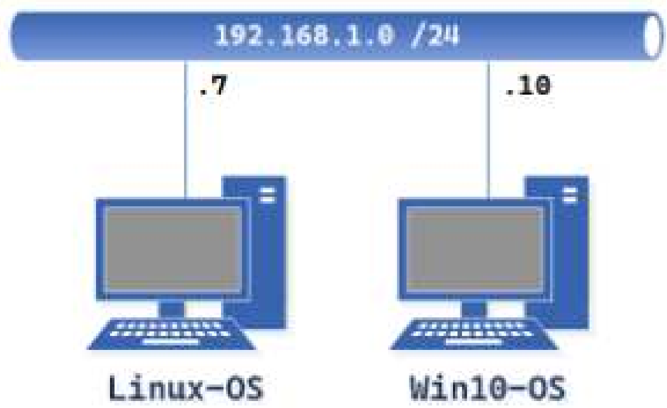
Objective

The tasks in this lab will explore the similarities and differences between Windows and Linux versions of network configuration and review key Linux network settings concepts.

In this lab, we will be performing the following tasks:

- Understand Static IP Configuration
- Familiarize yourself with IP Address Configuration Commands
- Configure and Manage File Sharing
- Utilize Network Command Line Utilities
- Troubleshoot and Verify Network Settings
- Contrast Windows and Linux Networking

Lab Topology



Lab Settings

The information in the table below will be needed to complete the lab. The task sections below provide details on the use of this information.

Virtual Machine	IP Address	Account (if needed)	Password (if needed)
Linux-OS	192.168.1.7 / 24	`sysadmin`	`NDGlabpass123!`
Win10-OS	192.168.1.10 / 24	`sysadmin`	`NDGlabpass123!`

1 Understanding Static IP Configuration

Understanding Static IP Configuration involves manually assigning a fixed IP address to a device, ensuring it remains consistent for reliable network services. This method contrasts with DHCP dynamic IP assignments, offering benefits such as stable connectivity for servers and other network devices. The process conceptually involves selecting an unused IP address on your network, configuring it with the necessary network details such as the subnet mask and gateway, either through a graphical user interface or the command line, and verifying the setup for conflicts or connectivity issues. This approach provides users with control and simplifies troubleshooting by maintaining a predictable network environment.

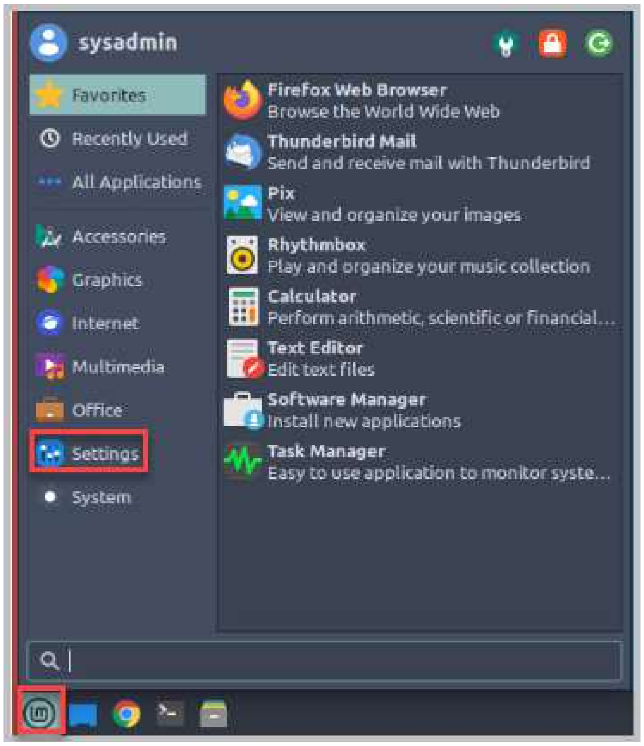
1.1 GUI IP Configuration with Linux

Setting a static IP on Linux via the GUI involves accessing network settings, selecting your connection type, manually inputting your IP details, and confirming the changes.

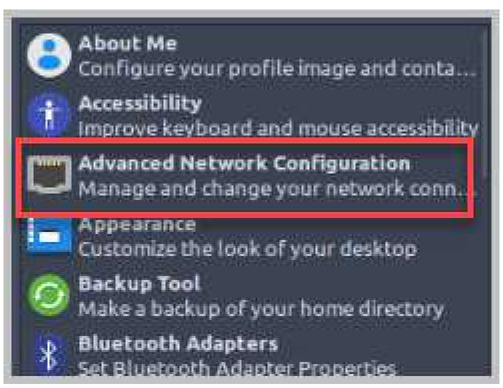
1. Open a console connection to the **Linux-OS** system.

> To launch the console for a lab device, you can click on the respective tab in the navigation bar at the top or navigate to the *Topology* section and click the lab device's corresponding image.

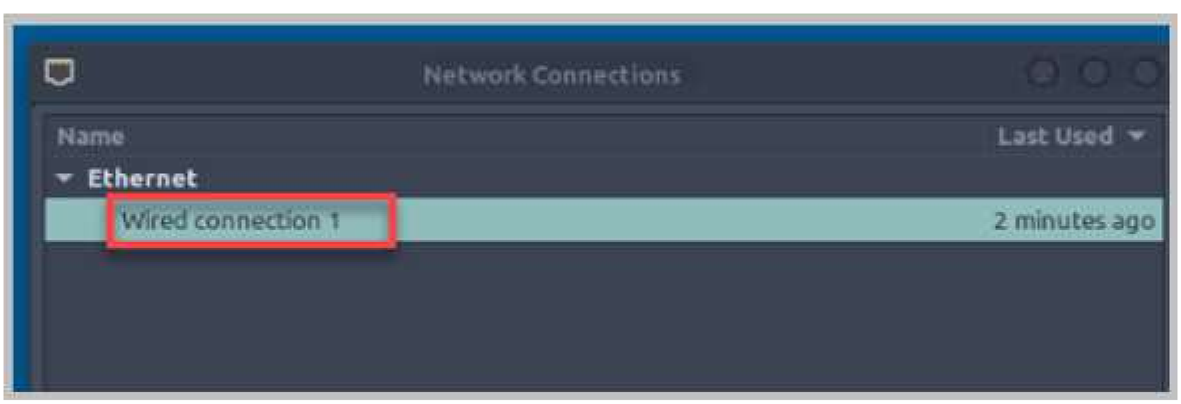
2. Select the **Start button** and select **Settings.**



3. In the list of settings, select **Advanced Network Configurations.**

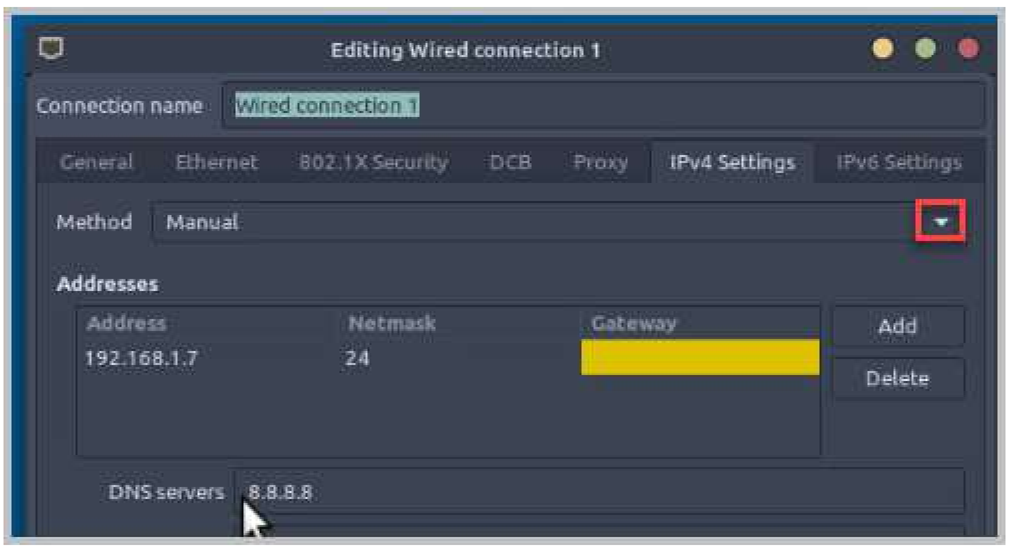


4. In the Network Connections window, double-click **Wired connection 1.**



5. Select the **IPv4 Settings** tab.

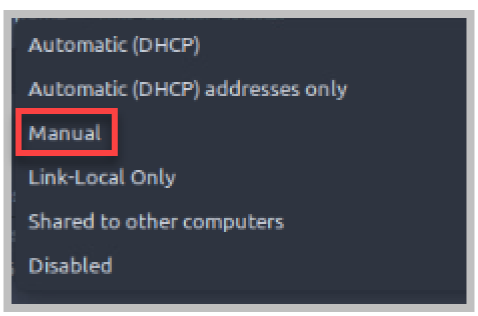
6. The method for receiving or setting a new IP address is already chosen by default. Currently, it's set to the Manual Method. Select the **drop-down arrow**.



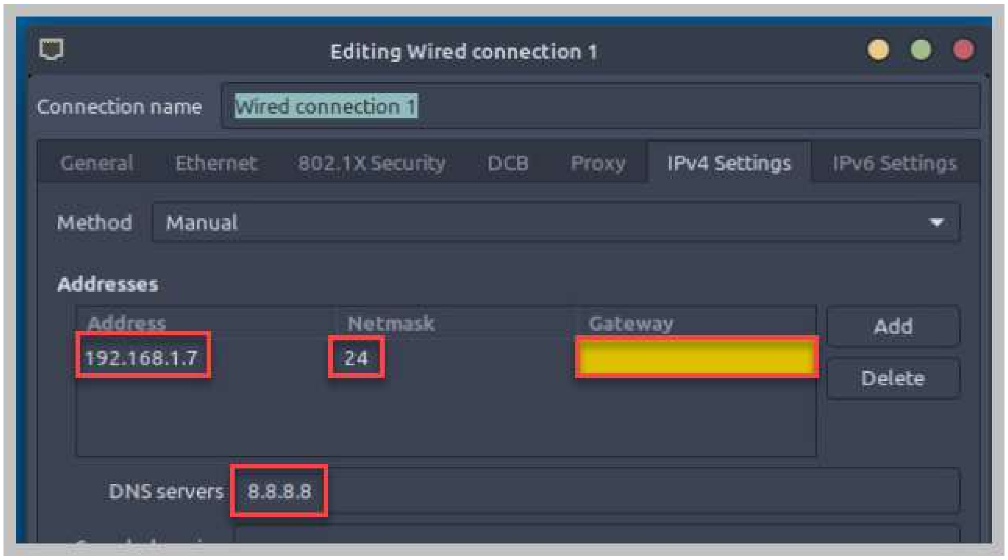
The list of available options allows users to tailor their network configuration to their specific needs, whether for ease of use with DHCP, precise control with manual settings, or special configurations like sharing connections or using link-local addresses for direct device communication.

- **Automatic (DHCP)**: The system automatically receives an IP configuration from a DHCP server on the network. This is the default for most home and office networks, where you don't need to manually set an IP.
- **Automatic (DHCP) Addresses Only**: Similar to DHCP, but allows you to manually specify DNS servers.
- **Manual**: Requires you to manually input all network details including IP address, netmask, gateway, and DNS servers. This is used when you need a static IP or in environments where DHCP isn't available.
- **Link-Local Only**: Uses a link-local address (like 169.254.x.x) for direct communication between devices on the same network segment without a router.
- **Shared to Other Computers**: Turns your computer into a router, sharing its internet connection with other devices. This is less common in modern setups but can be useful for creating ad-hoc networks.
- **Disabled**: Turns off the network interface, preventing it from connecting to any network.

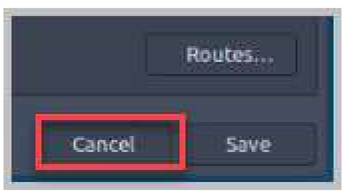
7. Select **Manual** to leave the existing static IP Configuration.



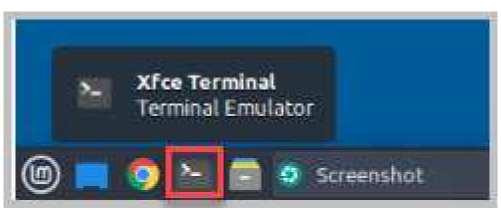
8. The **Manual** Network Method allows you to set the IP Configuration Details. Leave the existing settings. Verify the current IP Configuration settings.



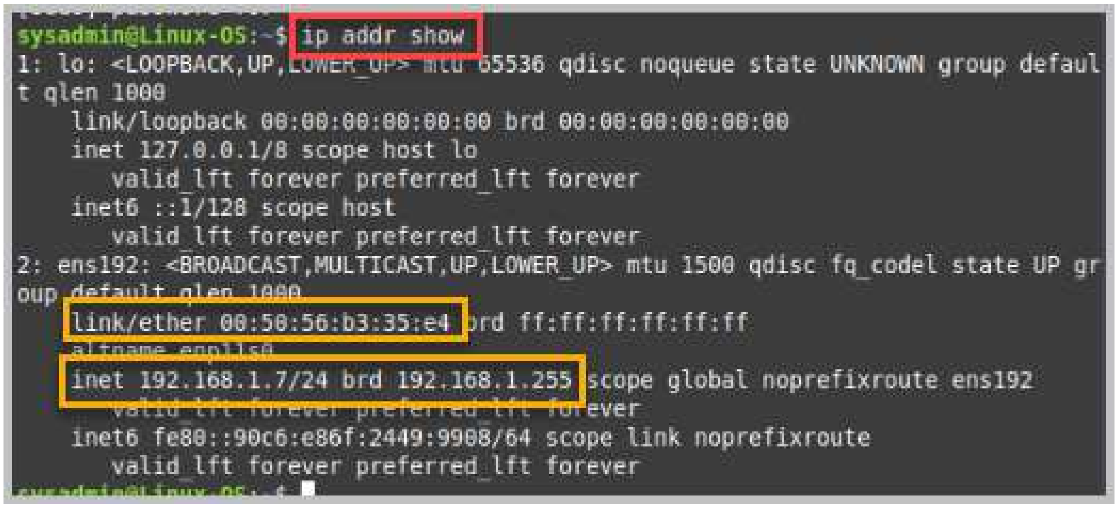
9. As no changes were made, select **Cancel.** If you had made any changes, you would have selected **Save** instead of **Cancel.**



10. Close the Network Connections window by selecting the **Red X** in the open window.
11. To verify the Static IP Configuration settings, select the **XFce Terminal Emulator** (Linux version of the Command Prompt).



12. In the terminal emulator window, type `ip addr show`. The **ip addr show** command lists all network interfaces along with their current configurations, including IP addresses, netmasks, and MAC addresses.

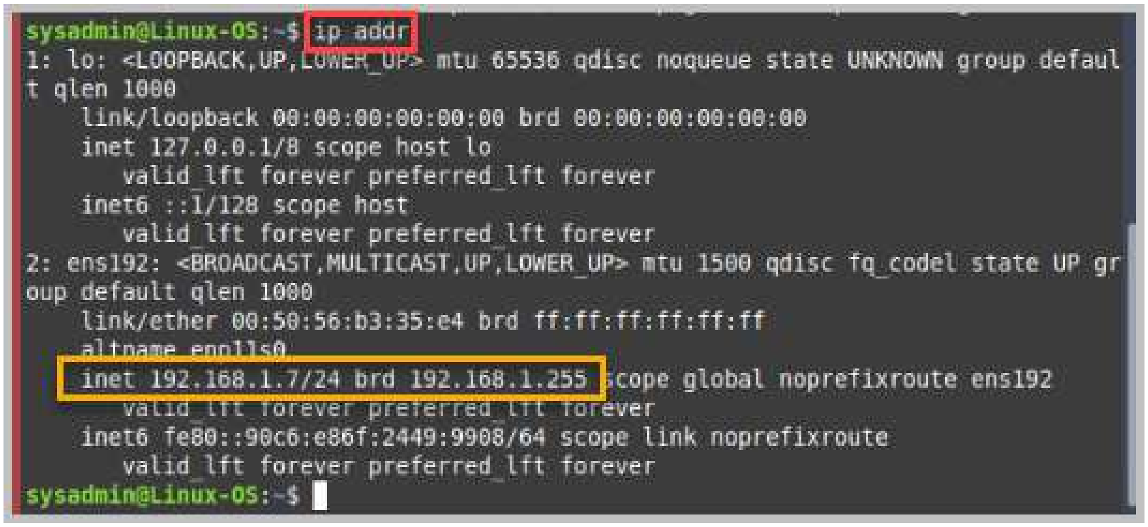


13. Leave the *Linux-OS* system with the *XFce Terminal Emulator (CLI)* opened to continue with the next task.

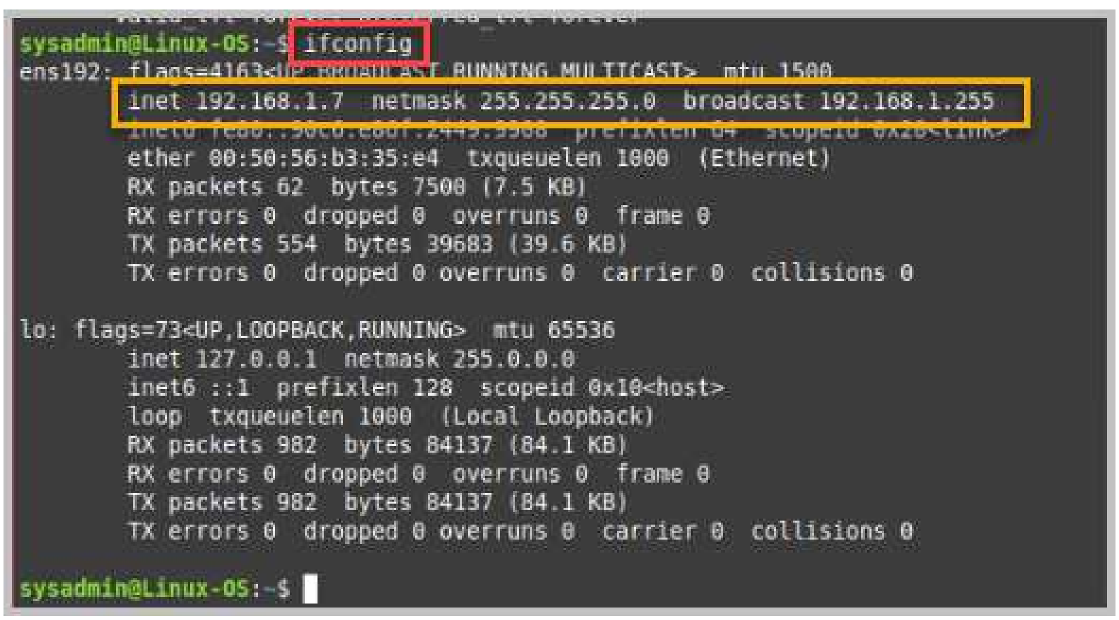
1.2 Static IP Configuration Verification Linux Commands

1. In the CLI (Command Line Interface), type `nmcli`. The **nmcli** command is a powerful tool for managing network connections on Linux systems using Network Manager. It provides a flexible and efficient way to control and monitor your network settings. Notice that this command includes the static IP Address, Subnet Mask (listed as /24), and DNS Server Address. To exit **nmcli**, type **Ctrl + C** on your keyboard.

2. In the CLI (Command Line Interface), type **ip addr.** This command displays your network interfaces and their IP configurations. Notice this command includes the static IP Address and Subnet Mask address listed as **/24,** but doesn't show the DNS Server Address.



3. In the CLI (Command Line Interface), type **ifconfig**. The **ifconfig** command is a utility for basic network interface configuration in Unix-like systems, showing or setting IP addresses and network status, though it's largely outdated by the **ip** command for modern network tasks. This command displays your network interfaces and their IP configurations. Notice that this command includes the static IP Address and Subnet Mask (listed as /24)**,** but no DNS Server Address.



> Using various commands to verify Linux network settings provides multiple benefits: it allows for cross-verification of network configurations, offers different perspectives on network state from kernel-level to service manager views, aids in troubleshooting by identifying discrepancies, ensures compatibility across different Linux environments, and deepens understanding of network layers. This approach enhances network management by offering redundancy, aiding in script automation, and providing educational insights into system architecture.

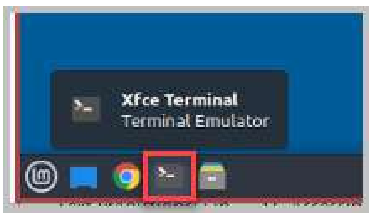
4. Close the *CLI* window. Leave the *Linux-OS* system open to continue with the next task.

2 Configure and Manage File Sharing

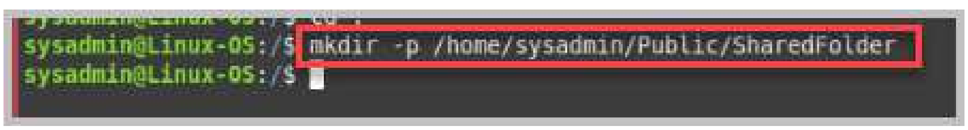
Shared folders and files in Linux offer numerous advantages. They foster collaboration by providing centralized access and enabling real-time teamwork. This streamlines workflows to boost productivity and improve communication. Additionally, shared resources optimize storage, reduce redundancy, and lower costs. Centralized security measures enhance data protection and reliable access. Common use cases include home networks, small offices, large organizations, and remote work environments.

2.1 Configure Network Sharing Settings

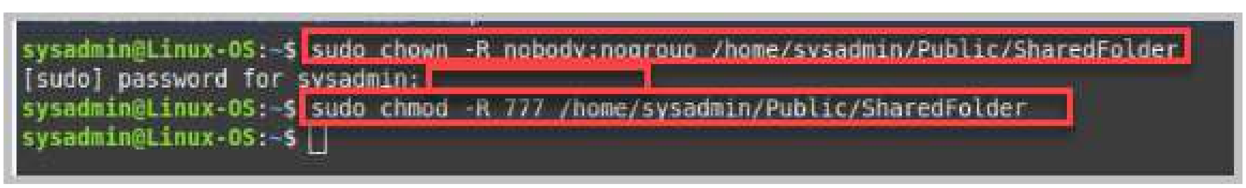
1. Open the **XFce Terminal** located on the taskbar.



2. Create a directory share by typing `mkdir -p /home/sysadmin/Public/SharedFolder`.



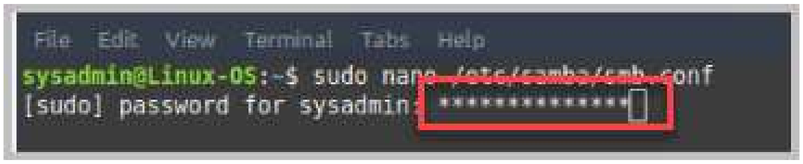
3. Set appropriate permissions for the shared folder by typing the following commands. After each command, hit the **Enter** key.
   - `sudo chown -R nobody:nogroup /home/sysadmin/Public/SharedFolder`
   - At prompt for sysadmin password, type `NDGlabpass123!`
   - Next type `sudo chmod -R 777 /home/sysadmin/Public/SharedFolder`



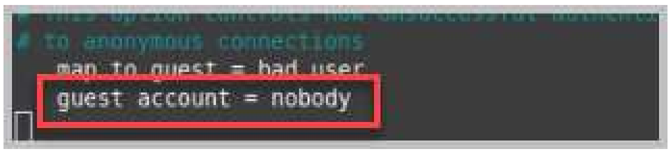
When setting up a shared directory for guest access, we assign nobody:nogroup as the owner of that directory. This allows anonymous users to access the share without requiring specific user credentials.

777 allows full access permissions for the SharedFolder.

4. Type `sudo nano /etc/samba/smb.conf` to open the Samba Configuration file. Enter the *sysadmin* password **NDGlabpass123!.**

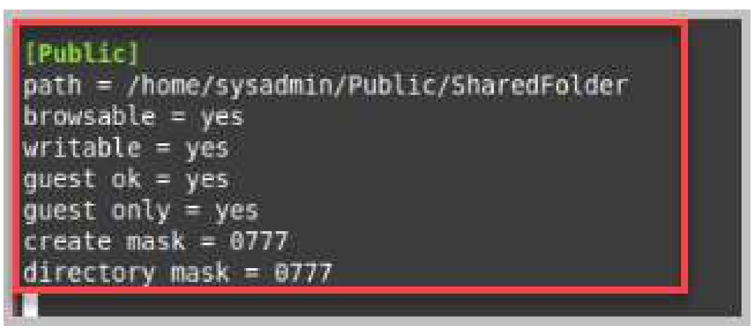


5. In the */etc/samba/smb.conf* scroll to the bottom of the file. Under the **Global Settings,** search for string *map to guest = Bad User,* select **Enter** on keyboard, and type **guest account = nobody**



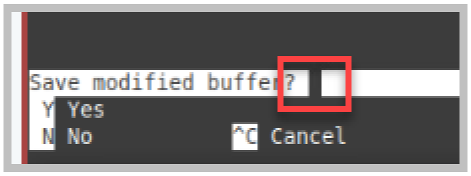
6. Scroll to the bottom of the */etc/samba/smb.conf* file and type the following commands. After each command, hit the **Enter** key on your keyboard.

```
[Public]
path = /home/sysadmin/Public/SharedFolder
browsable = yes
writable = yes
guest ok = yes
guest only = yes
create mask = 0777
directory mask = 0777
```



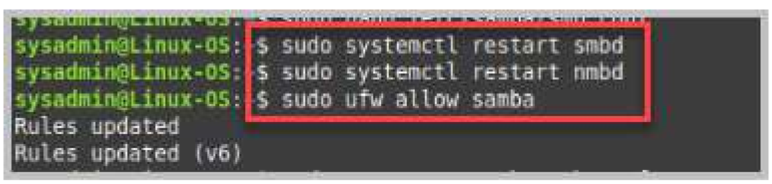
- guest account = nobody: Specifies that the guest account used for anonymous access is nobody.
- browsable = yes: Makes the share visible when browsing the network.
- writable = yes: Allows write access to the folder.
- guest ok = yes: Allows guest access.
- guest only = yes: Ensures only guest access is allowed, so no username/password prompt occurs.
- create mask and directory mask: Sets permissions for files and directories created in the share.

7. Once you have verified your input, hit **Ctrl + X** on your keyboard. When prompted to Save Modified Buffer, type **Y.**



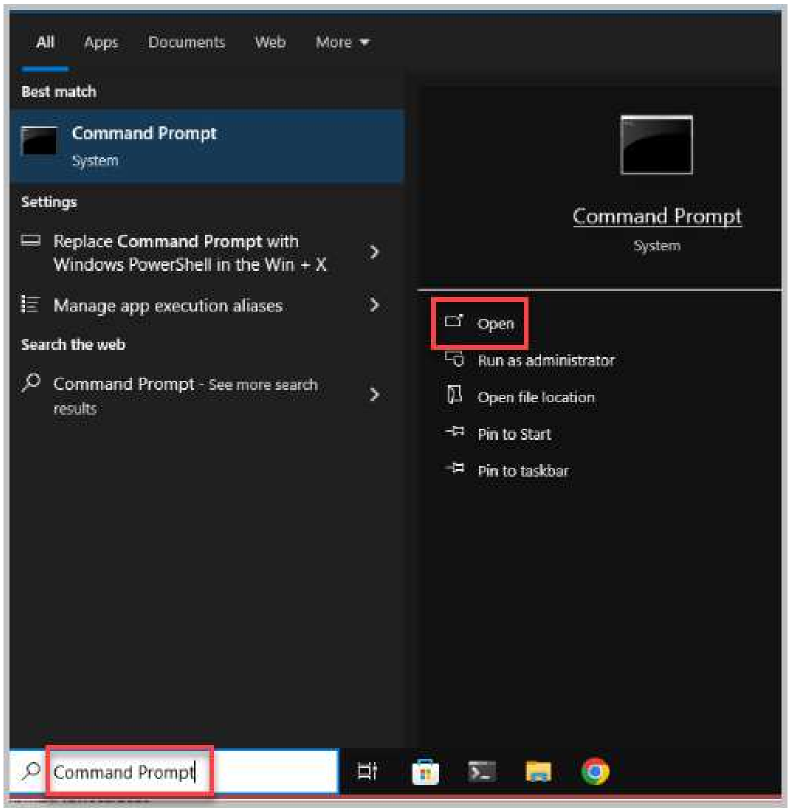
8. The name and location of the *Samba* file is shown. Hit **Enter** to accept the name and close the nano editor.

9. Samba Services needs restarting. Type **`sudo systemctl restart smbd`,** hit **Enter** on the keyboard. Type **`sudo systemctl restart nmbd`,** hit **enter** on the keyboard. To ensure the firewall allows Samba traffic, type **`sudo ufw allow samba`.**

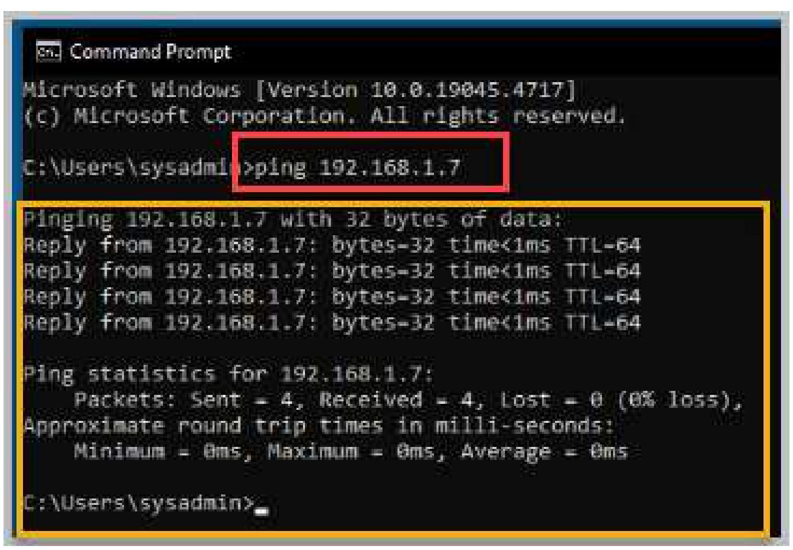


10. Close all open windows, including the XFce Terminal.
11. Open a console connection to the **Win10-OS** system.

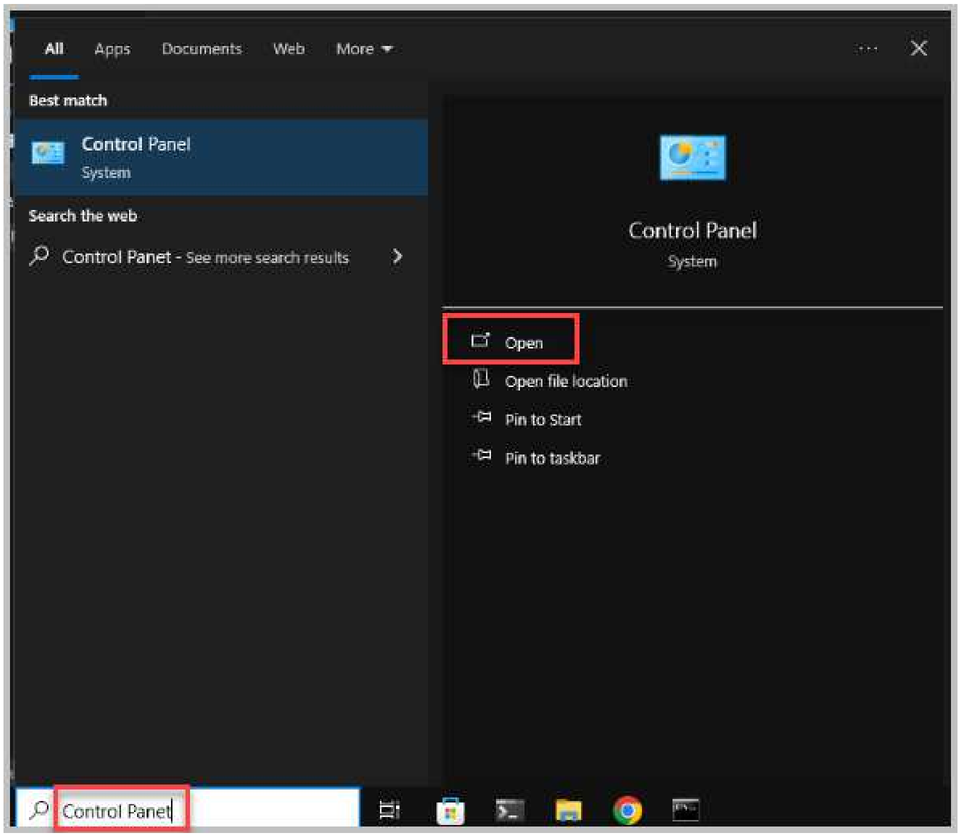
12. In the search bar on the taskbar, type **Command Prompt,** and select **Open.**



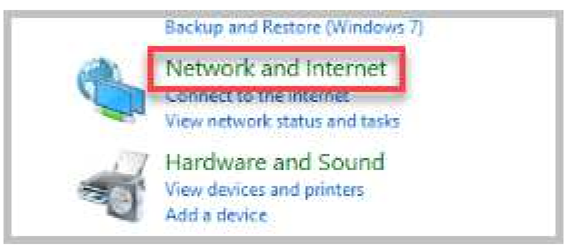
13. Ping the Linux-OS system IP Address by typing **`ping 192.168.1.7`** and selecting **Enter.** The screenshot below shows a successful ping from the *Win10-OS System to the Linux-OS System.*



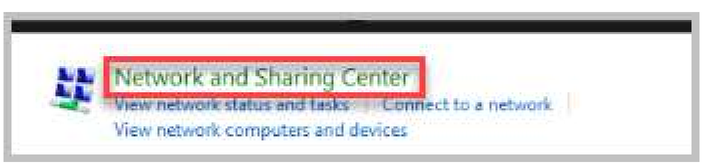
14. In the search box in the taskbar, type **Control Panel.** Select **Open.**



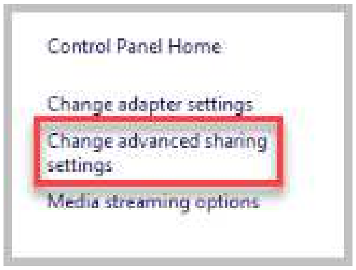
15. Select **Network and Internet.**



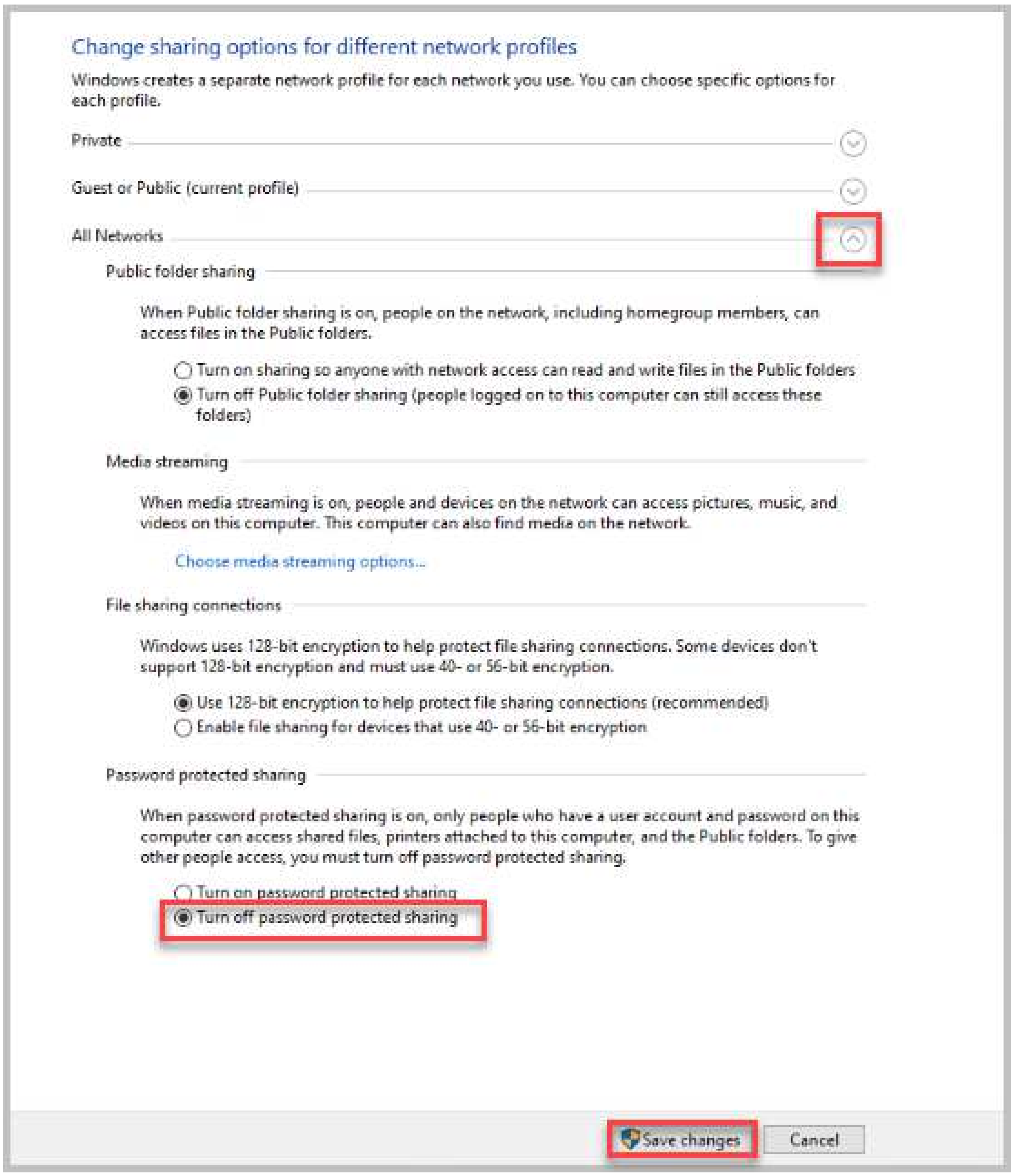
16. Select **Network and Sharing Center.**



17. Select **Change advanced sharing settings.**

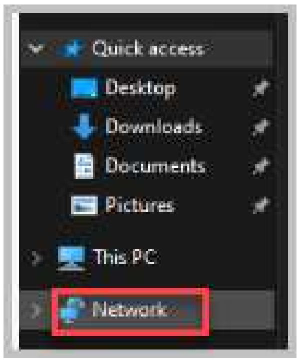


18. Select the **drop-down arrow** for All Networks. Select **Turn off password protected sharing.** Select **Save changes.** Close all open windows.



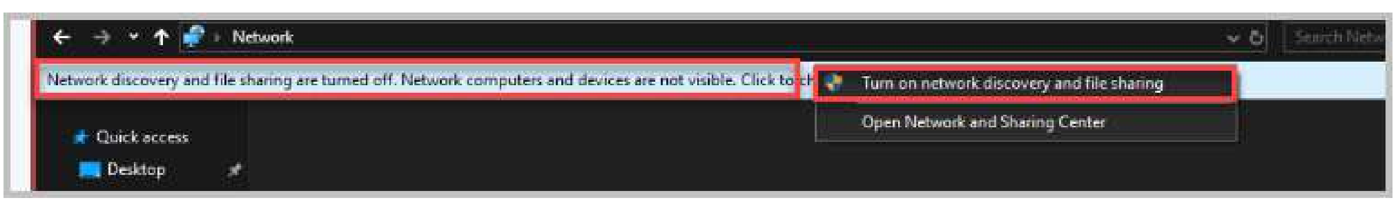
19. Select **File Explorer** in the taskbar.

20. Under Quick Access of File Explorer, select the **Network**.

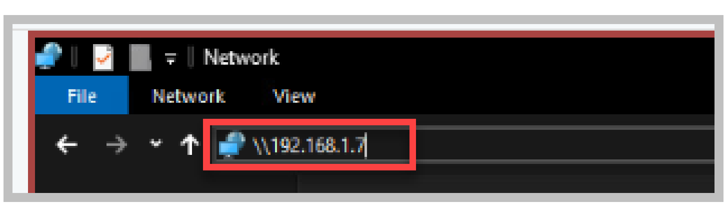


21. Select **Ok,** acknowledging that *Network Discovery is turned off.*

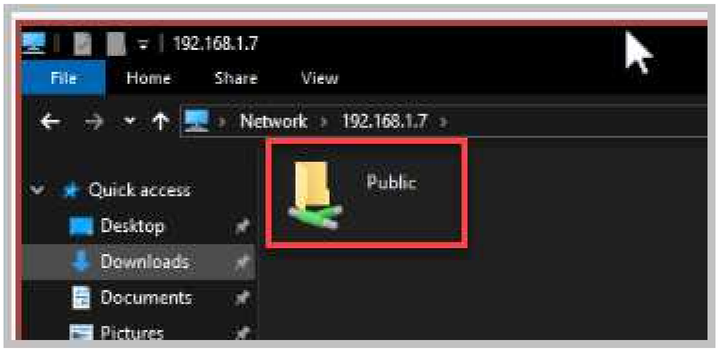
22. Select the yellow notification. *Network discovery and file sharing are turned off. Network computers and devices are not visible.* **Click** to change. Select **Turn on network discovery and file sharing.**



23. In the *Address Bar,* type `\\192.168.1.7` and hit **Enter.**



24. A successful connection should occur as the *Public* folder is now showing correctly.



25. Close all open windows. Continue to the next activity.

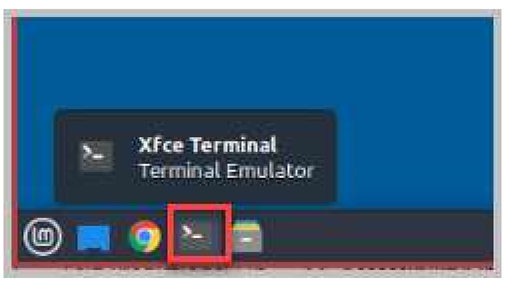
3 Using Network Command-Line Utilities

All the computers you are working on in the lab topology are on the same subnet and connected to the network. Hardware issues, firewall settings, incorrect IP addressing, and errors in network configuration can all cause problems with network connectivity. In this task, you will be using command-line tools to view useful information for discovering network configurations and troubleshooting.

3.1 Interacting with the Linux-OS CLI

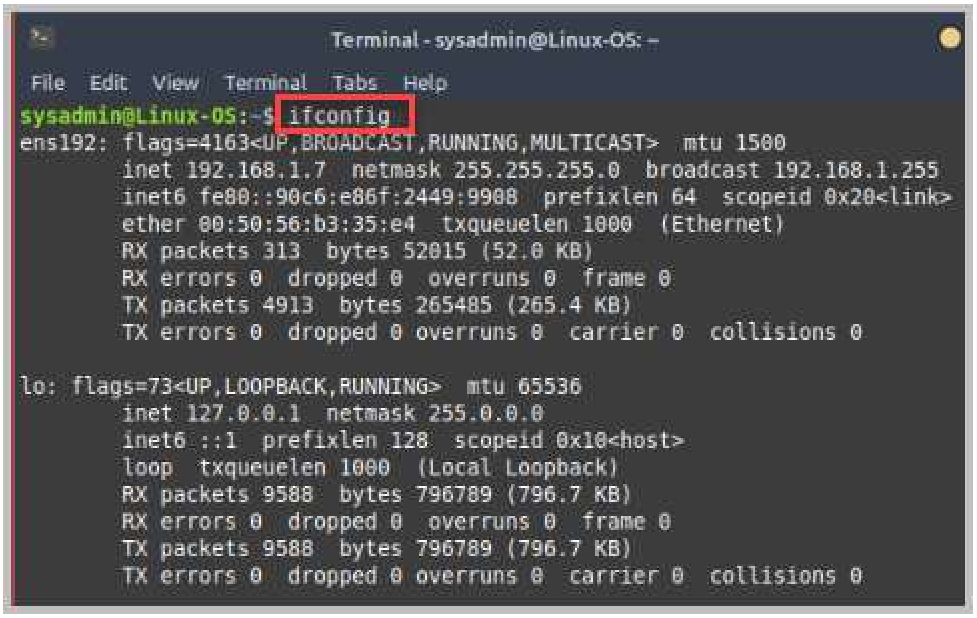
1. Open a console connection to the **Linux-OS** system.

2. Open the **XFce Terminal** located on the taskbar.



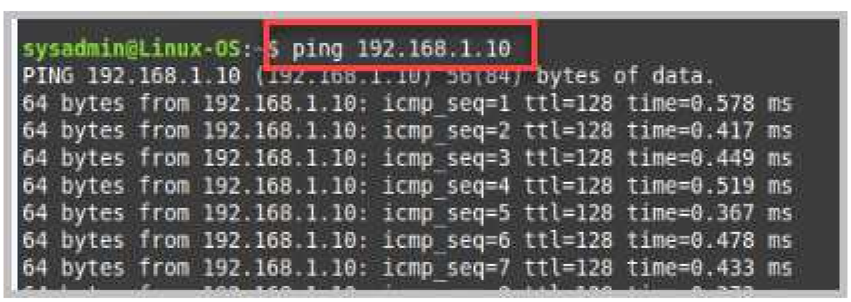
3. In the command prompt window, type the `ifconfig` command and press **Enter** to view help for the command. This command will list all network interfaces and their basic configuration.

```
ifconfig
```



4. In the command prompt interface, enter the **`ping 192.168.1.10`** command. Is it successful? When referring to the topology, this is the *IP* address of the *Win10-OS* host. Using the *ping* command verifies connectivity to it. The *ping* command will continue until you type **Ctrl + C** on the keyboard.

```
ping 192.168.1.10
```



5. At the command prompt, enter the **`tracepath 192.168.1.10`** command. *Tracepath* traces the route a packet takes to reach a destination and shows information about each hop along that route. Based on the command output, this device is on the same local subnet as *Win10-OS.* To stop the *tracepath* command, press **Ctrl + C**.

```
tracepath 192.168.1.10
```

Because this lab environment is a flat network (a single subnet), only one hop is shown.

6. Enter the **`netstat --help`** command and review the available options. This network utility displays network connections from the host machine. Try this command with any of the options below.

```
netstat --help
```

```
sysadmin@Linux-OS:~$ netstat --help
usage: netstat [-vWeenNcCF] [<Af>] -r         netstat {-V|--version|-h|--help}
       netstat [-vWnNcaeol] [<Socket> ...]
       netstat { [-vWeenNac] -i | [-cnNe] -M | -s [-6tuw] }

        -r, --route              display routing table
        -i, --interfaces         display interface table
        -g, --groups             display multicast group memberships
        -s, --statistics         display networking statistics (like SNMP)
        -M, --masquerade         display masqueraded connections

        -v, --verbose            be verbose
        -W, --wide               don't truncate IP addresses
        -n, --numeric            don't resolve names
        --numeric-hosts          don't resolve host names
        --numeric-ports          don't resolve port names
        --numeric-users          don't resolve user names
        -N, --symbolic           resolve hardware names
        -e, --extend             display other/more information
        -p, --programs           display PID/Program name for sockets
        -o, --timers             display timers
        -c, --continuous         continuous listing

        -l, --listening          display listening server sockets
        -a, --all                display all sockets (default: connected)
        -F, --fib                display Forwarding Information Base (default)
        -C, --cache              display routing cache instead of FIB
        -Z, --context            display SELinux security context for sockets

  <Socket>={-t|--tcp} {-u|--udp} {-U|--udplite} {-S|--sctp} {-w|--raw}
           {-x|--unix} --ax25 --ipx --netrom
  <AF>=Use '-6|-4' or '-A <af>' or '--<af>'; default: inet
  List of possible address families (which support routing):
    inet (DARPA Internet) inet6 (IPv6) ax25 (AMPR AX.25)
    netrom (AMPR NET/ROM) ipx (Novell IPX) ddp (Appletalk DDP)
    x25 (CCITT X.25)
sysadmin@Linux-OS:~$
```

7. *Nslookup* is a command-line utility that queries the DNS to obtain a domain name, IP address mapping, or other types of DNS records. Enter the **`nslookup netdevgroup.com`** command.

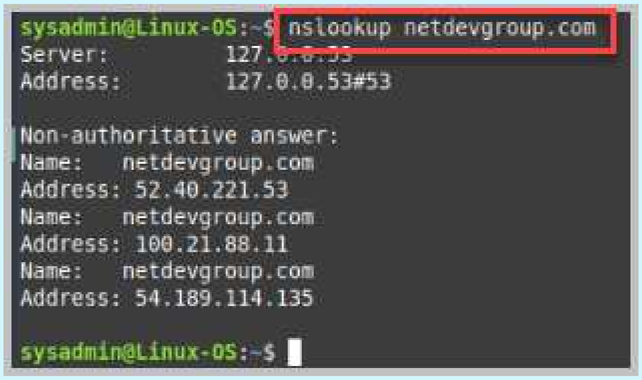
```
nslookup netdevgroup.com
```

```
sysadmin@Linux-OS:~$ nslookup netdevgroup.com
;; Got SERVFAIL reply from 127.0.0.53
Server:         127.0.0.53
Address:        127.0.0.53#53

** server can't find netdevgroup.com: SERVFAIL

sysadmin@Linux-OS:~$
```

Example output of a Linux system using DNS server.



In this lab environment, which is not connected to the internet and lacks a DNS server, Linux systems are unable to resolve domain names through DNS. By default, tools like *nslookup* on Linux will attempt to query a DNS server for records. However, if no DNS server is configured or reachable, *nslookup* will not resolve names.

8. The lab is now complete; you may end the reservation.